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Tab 1

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2

Research Topic: *SLC6A4 DNA Methylation as a Potential Biological Link Between Childhood Adversity and Reward-Learning During Adolescence*

SLC6A4 DNA Methylation as a Potential Biological Link Between Childhood Adversity and Reward-Learning During Adolescence

Abstract: Childhood adversity has been researched and shown to be associated with altered reward-learning. Increased exposure to adversity may influence adolescent reward-learning by increasing sensitivity to negative feedback. However, gaps in existing research remain regarding the biological mechanisms underlying the relationship between adversity and altered reward-learning. One potential pathway through which adversity may influence learning is epigenetic regulation. Environmental stressors have been associated with changes in DNA methylation, which can influence gene expression. *SLC6A4*, a gene encoding the serotonin transporter, may represent a potential biological link between adversity and altered reward-learning. This study investigates *SLC6A4* DNA methylation as a potential link and hypothesizes that childhood adversity-associated *SLC6A4* DNA methylation will be associated with altered reward-learning in adolescents. The study involves an in-depth literature review and a proposed mixed-method methodology involving adolescent participants. The proposed study will assess *SLC6A4* DNA methylation, identify varying levels of childhood adversity, and measure reward-learning using a probabilistic behavioral task with a reversal component. The findings could contribute to future research in psychology, biology, and neuroscience by examining whether adversity-associated epigenetic differences are reflected in behavioral manifestations of reward-learning and decision-making during adolescence.

Introduction: Adversity is broadly defined as exposure to distressing or harmful life circumstances, ranging from abuse to neglect. In this study, adversity specifically refers to early-life stress (ELS) associated with unstable and abusive caregiving environments, including emotional and physical abuse such as hitting or slapping, excessive manipulation or degradation, controlling behavior, and sexual abuse. Childhood adversity has been associated with differences in how individuals learn from rewards, risks, and feedback. Reward-learning involves an individual's ability to recognize signals in the environment and associate them with rewards or risks. This process contributes to decision-making by allowing individuals to modify future behavior based on positive or negative feedback [1]. Reward-learning is particularly significant during adolescence, a period in which individuals develop behavioral strategies for responding to their environment, including experiences of hardship and success. Childhood experiences make up building blocks that form a child's mental framework and basis of future stress response and action. Importantly, adverse experiences do not necessarily prevent individuals from recognizing associations between events and rewarding outcomes. Instead, adversity may alter how individuals assign motivational value to this information and incorporate it into future behavioral decisions.

Although the relationship between adversity and altered reward-learning has been increasingly studied, the mechanisms beneath this relationship remain not entirely understood. Children exposed to adverse experiences may recognize standard positive outcomes but prioritize avoidance of potential risks, devalue information about potential rewards, present greater sensitivity to negative than positive feedback, demonstrate increased sensitivity to reversals from positive to negative outcomes than from negative to positive outcomes, and exhibit altered activity in reward-related neural regions during reward anticipation and reception [1]. These findings demonstrate associations between adversity and differences in reward-related

behavior and neural processing; however, the biological pathways underlying these differences require further investigation. One proposed mechanism through which environmental experiences may become biologically embedded is through epigenetic changes.

Literature Review: Epigenetics refers to biological mechanisms that regulate gene expression without changing the underlying DNA sequence. One mechanism is DNA methylation, which could potentially contribute to the biological effects of adversity on behavior. DNA methylation occurs when a methyl group is added to cytosine-guanine sites on DNA, known as CpG sites [2]. Depending on its location and biological context, DNA methylation can influence whether a gene is expressed. For example, methylation within a promoter region can interfere with transcription by recruiting proteins that repress gene expression [2].

This mechanism is relevant because specific methylation patterns have been associated with early-life social experiences. For example, family psychosocial adversity has been associated with increased or decreased DNA methylation at distinct sets of genomic CpG sites [3]. Studies examining peripheral DNA from blood or saliva in infants, adolescents, and adults have also reported associations between childhood abuse or neglect and increased *NR3C1* methylation. The *NR3C1* gene is involved in regulation of the body's stress response, providing evidence that epigenetic regulation may represent one biological mechanism through which experiences of adversity become biologically embedded [4].

This raises the question of which specific genes may be associated with adversity and could also be relevant to reward-learning. One potential candidate is *SLC6A4*.

The *SLC6A4* gene encodes the serotonin transporter, which is responsible for the reuptake of serotonin from the synaptic space into presynaptic neurons [5]. Serotonin signaling plays an important role in regulating mood, emotion, motivation, and behavior. Serotonergic neurons have also been associated with reward-learning [6]. Research has shown that serotonergic neurons can respond to punishment and that a subset of these neurons is activated by reward-predicting cues, with different patterns of signaling occurring across different stages of reward and punishment processing. These findings provide evidence that serotonergic signaling is involved in processing information about rewards and punishments.

Because *SLC6A4* regulates serotonin reuptake [7], changes in *SLC6A4* expression could potentially alter serotonergic signaling and, in turn, influence reward-related processes. However, this does not establish that *SLC6A4* methylation directly causes changes in reward-learning. Instead, it provides a biological reason for investigating whether the two processes are associated.

SLC6A4 methylation has also emerged as a potential marker of adversity exposure [8]. Research on adversity and DNA methylation, including findings involving *NR3C1*, demonstrates that environmental experiences can be associated with differences in gene regulation. Regarding *SLC6A4*, altered methylation is relevant because serotonin influences numerous behavioral processes associated with reward and motivation, including, for example, impulsivity and goal-directed behavior. Research involving serotonergic manipulation has further demonstrated effects on goal-directed versus habitual behavior, while several serotonin receptor subtypes,

including 1A, 1B, 2A, 2C, and 3, have also been involved in reward-related behavior and reward valuation [9], [10], [11], [12].

These findings provide a basis for investigating whether adversity-associated *SLC6A4* DNA methylation may be related to altered reward-learning during adolescence. Although childhood adversity has been associated independently with altered reward-learning and changes in DNA methylation, and additionally *SLC6A4* methylation has been looked into as a potential marker of adversity exposure, the potential relationship and pathway between adversity-associated *SLC6A4* methylation and altered reward-learning during adolescence remains in question.

Research question & Hypothesis: A clear and directed research question was developed through preliminary research. Existing studies have identified associations between childhood adversity and altered reward-learning, as well as between adversity and epigenetic changes. However, the potential relationship between adversity-associated epigenetic changes and altered reward-learning remains insufficiently characterized. The proposed research question is:

Could *SLC6A4* DNA methylation represent a biological pathway linking childhood adversity to altered reward-learning during adolescence?

Hypothesis: Childhood adversity-associated *SLC6A4* DNA methylation will be associated with altered reward-learning during adolescence.

This hypothesis examines whether *SLC6A4* DNA methylation may represent a biological pathway connecting an environmental experience with differences in reward-related learning during adolescence.

Proposed Study & Methodology: The proposed research study would follow a qualitative and quantitative framework with a large sample size and standardized, repeatable structure. The study would be conducted according to 3 primary study variables/components:

Childhood Adversity: Adolescents aged 12-17 would be subjected to questioning to assess level of adversity. The selected age range encompasses a sensitive window of neural development at which regions such as the prefrontal cortex undergo maturation and significant changes—changes that adversity can possibly be responsible for altering. Questions would assess varying levels, types, and chronicity of adversity. Participants would be categorized into three groups based on the severity and characteristics of reported adversity. These categories would be determined for research purposes and would not be disclosed to participants, avoiding the assignment of possibly stigmatizing labels. The questionnaire would provide a quantitative measure of adversity that could be compared with biological and behavioral measures.

***SLC6A4* methylation:** Participants would provide a saliva or blood sample to assess DNA methylation at selected CpG sites within the *SLC6A4* gene. A methylation assay, such as bisulfite pyrosequencing, would be used to treat genomic DNA and quantify methylation at the selected sites. This would provide a quantitative measure of the percentage of DNA methylation at the targeted regions. Peripheral methylation should be interpreted cautiously, as methylation measured in blood or saliva is not a direct measure of methylation throughout the brain.

Reward-Learning: Reward-learning is a behavioral variable. Reward-learning would be measured using a validated probabilistic reinforcement-learning task consisting of repeated trials in which participants choose between two symbols and learn through trial and error which symbol has a higher probability of producing a reward. After each choice, the participant will be provided either a *reward*—positive feedback—or a *punishment*—negative feedback. Participants are expected to learn overtime which choice gives them more rewards. The study will also include a reversal phase where previously learned reward circumstances are changed, analyzing how swiftly a participant can *adjust* their prior notion and expectations. This test uses a model of positive and negative feedback to test individuals' ability to learn reward-outcome associations and update those associations following changes in outcome based on:

- Learning rate -> how quickly behavior is changed after feedback
- Reward sensitivity -> how strongly positive feedback influences subsequent choice
- Punishment sensitivity -> how strongly negative feedback influences subsequent choice
- Reversal learning -> what happens when the previously rewarded choice stops being rewarded
- Choice accuracy -> whether they increasingly choose the option associated with reward

Following the behavioral task, participants would complete a short questionnaire about their decisions. This questionnaire would provide qualitative insight into the reasoning behind their choices and could help contextualize unexpected responses or outliers in the behavioral data. The qualitative responses would supplement the quantitative measures obtained from the reward-learning task.

Together, these three components would provide a combination of biological, behavioral, and self-reported data for examining the relationship between childhood adversity, *SLC6A4* methylation, and reward-learning. Following data collection, associations among these measures would be analyzed to determine whether greater childhood adversity is associated with differences in *SLC6A4* methylation and whether these methylation differences are associated with altered reward-learning performance.

Relevance

Scientific: This study addresses the intersection of epigenetics, gene regulation, and the biological effects of childhood adversity. By examining *SLC6A4* DNA methylation alongside reward-learning behavior in adolescents, the proposed research could provide insight into whether epigenetic variation may be associated with differences in reward-related learning following experiences of adversity. The findings could contribute to further investigation of how environmental experiences may become biologically embedded and relate to behavioral development.

Clinical Research: The proposed study focuses on the *SLC6A4* gene, which encodes the serotonin transporter and plays an important role in serotonin regulation [5]. Understanding the relationship between adversity, *SLC6A4* methylation, and reward-learning may contribute to broader research on emotional regulation, reward-related behavior, and serotonergic function. Because the serotonin transporter is also a major target of selective serotonin reuptake inhibitors (SSRIs), research examining *SLC6A4* and its regulation may potentially have relevance to future

research on serotonergic treatments [5], [10]. However, the proposed study would not specifically evaluate SSRI effectiveness or treatment outcomes.

Psychological: The study examines adolescents who have experienced varying levels of childhood adversity and evaluates whether differences in adversity are associated with differences in reward-learning behavior. By combining self-reported adversity, a biological measure of *SLC6A4* methylation, and behavioral measures of reward-learning, the study seeks to connect environmental experiences with biological and psychological processes involved in decision-making. The findings could help clarify potential pathways connecting childhood adversity with differences in reward-related behavior during adolescence.

Limitations & Project Logistics: This project would require informed consent and signatures from both adolescents and their parents or legal guardians, creating additional concerns associated with conducting research involving minors. The study would require a considerably large sample size to examine associations among childhood adversity, *SLC6A4* methylation, and reward-learning while accounting for variation in age and demographic characteristics. However, obtaining a large and diverse adolescent sample may present other challenges.

The study would involve three primary phases: assessment of childhood adversity, collection and analysis of biological samples for *SLC6A4* methylation, and behavioral assessment of reward-learning. The biological aspect would require accessibility to appropriate laboratory facilities and individuals capable of conducting methylation assays and properly managing biological and participant data.

The study would also be subject to limitations associated with self-report. Participants may lie, interpret questions differently, or may be unwilling to fully report sensitive experiences, creating the possibility of response or recall bias. Because the study is observational, associations between adversity, *SLC6A4* methylation, and reward-learning would not necessarily establish causation. Additionally, methylation measured from peripheral samples such as saliva or blood would not directly represent the exact type of methylation occurring in the brain. These limitations would need to be considered when interpreting the findings.

Ethical Implications: Since the proposed study involves minors and questions about potentially sensitive experiences of childhood adversity, participant safety and privacy would be essential. Participation would require informed consent from a parent or legal guardian as well as consent from the adolescent. Participants would be informed that participation is in fact voluntary and that they may decline to answer some questions or withdraw from the study.

Personal data and biological information would have to be kept confidential and stored using appropriate security procedures. Researchers would also need to determine appropriate procedures prior to the study for responding to disclosures of current abuse or immediate safety concerns, aligning procedures with applicable research and mandatory-reporting requirements.

The reward-learning task and adversity questionnaire would be designed to minimize unnecessary stress. Questions will be carefully written and corroborated by psychology professionals including both researchers and psychologists. Participants will also be provided

with appropriate support or referral information if participation raises any emotional concerns. Because the study involves biological samples, participants and guardians would also need to be informed on what is being collected, how samples would be analyzed, how long data would be retained, and how biological information would be protected and used.

Conclusion: In conclusion, this research study investigates associations among childhood adversity, *SLC6A4* methylation and reward-learning in adolescents. The research takes a mixed-method approach, including qualitative and quantitative data as well as self-report techniques and DNA analysis. By examining these measures together, the study seeks to determine whether differences in *SLC6A4* methylation are associated with the relationship between childhood adversity and altered reward-learning.

The study is expected to find differences in both reward-learning and gene methylation correlated with varying levels of adversity, filling an evident gap in mechanism and pathway research of *SLC6A4* methylation, its biological connection to decision-making and adaptability, and adversity's effects on both subject areas. In particular, the study could provide a basis for further investigation into the relationship between *SLC6A4* regulation, serotonin-related processes, and decision-making.

This research has potential relevance to biological, psychological, and clinical research. It could identify patterns and associations that inform future studies examining how environmental experiences may become biologically embedded and relate to behavioral development. Although the involvement of adolescent participants creates ethical and privacy concerns, the proposed methods to mitigate these concerns, including informed consent, confidentiality, and procedures for addressing potential distress or safety concerns, are present to help address participant safety and emotional stability.

Timeline

Week 1: Literature review — Review existing research on childhood adversity, gene methylation, and reward-learning to create a strong informational foundation for the study.

Week 2: Study design and preparation — Plan and finalize the participant criteria, the adversity level assessment along with conducted expert review, the *SLC6A4* methylation analysis technique and material accessibility, and the reward-learning task and functionality.

Week 3: Data collection — Recruit adolescent participants, extract DNA samples for methylation analysis, introduce privacy safeguards, collect necessary consent waivers, and administer reward-learning tests and adversity questionnaires.

Week 4: Data analysis — Organize data, categorize qualitative and quantitative findings, and analyze associations from data on childhood adversity, *SLC6A4* methylation, and observed reward-learning.

Week 5: Study write up and conclusion — Consider study limitations, synthesize and interpret findings, evaluate proposed research question and hypothesis, finalize research paper consolidating discoveries, and develop recommendations for future research.

Post Project Plan

Following the proposed research study, future research should further investigate:

- Biological pathways connecting *SLC6A4* gene expression, serotonergic function, and reward-learning
- Detailed psychological implications of altered *SLC6A4* gene expression
- Relationship between childhood adversity and altered reward-learning

To further clarify relationships, future studies should be conducted analyzing individual connections between reward-learning, *SLC6A4* methylation, and childhood adversity. Separating each relationship into its own distinct and focused study will allow for potential mechanisms to be researched and characterized in great detail prior to integrating findings into a broader, more defined and informed pathway. Future research should also utilize larger and more diverse samples. It should additionally incorporate direct measures of serotonergic function and serotonin production to investigate if the associations shown in this study could represent a broader, more connected mechanism.

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Tab 2

SLC6A4 DNA Methylation as a Potential Biological Link Between Childhood Adversity and Reward-Learning During Adolescence

Abstract

Childhood adversity has been researched and shown to be associated with altered reward-learning. Increased exposure to adversity may influence adolescent reward-learning by increasing sensitivity to negative feedback. However, gaps in existing research remain regarding the biological mechanisms underlying the relationship between adversity and altered reward-learning. One potential pathway through which adversity may influence learning is epigenetic regulation. Environmental stressors have been associated with changes in DNA methylation, which can influence gene expression. *SLC6A4*, a gene encoding the serotonin transporter, may represent a potential biological link between adversity and altered reward-learning. This study investigates *SLC6A4* DNA methylation as a potential link and hypothesizes that childhood adversity-associated *SLC6A4* DNA methylation will be associated with altered reward-learning in adolescents. The proposed study will assess *SLC6A4* DNA methylation, identify varying levels of childhood adversity, and measure reward-learning using a probabilistic behavioral task with a reversal component.

Introduction

Adversity is broadly defined as exposure to distressing or harmful life circumstances, ranging from abuse to neglect. In this study, adversity specifically refers to early-life stress (ELS) associated with unstable and abusive caregiving environments. Childhood adversity has been associated with differences in how individuals learn from rewards, risks, and feedback. Reward-learning involves an individual's ability to recognize signals in the environment and associate them with rewards or risks. This process contributes to decision-making by allowing individuals to modify future behavior based on positive or negative feedback [1]. Reward-learning is particularly significant during adolescence, a period in which individuals develop behavioral strategies for responding to their environment, including experiences of hardship and success. Importantly, adverse experiences do not necessarily prevent individuals from recognizing associations between events and rewarding outcomes. Instead, adversity may alter how individuals assign motivational value to this information and incorporate it into future behavioral decisions.

Although the relationship between adversity and altered reward-learning has been increasingly studied, the mechanisms beneath this relationship remain not entirely understood. Children exposed to adverse experiences may recognize standard positive outcomes but prioritize avoidance of potential risks, devalue information about potential rewards, present greater

sensitivity to negative than positive feedback, demonstrate increased sensitivity to reversals from positive to negative outcomes than from negative to positive outcomes, and exhibit altered activity in reward-related neural regions during reward anticipation and reception [1]. One proposed mechanism through which environmental experiences may become biologically embedded is through epigenetic changes.

Literature Review

Epigenetics refers to biological mechanisms that regulate gene expression without changing the underlying DNA sequence. One mechanism is DNA methylation, which could potentially contribute to the biological effects of adversity on behavior. DNA methylation occurs when a methyl group is added to cytosine-guanine sites on DNA, known as CpG sites [2]. Depending on its location and biological context, DNA methylation can influence whether a gene is expressed [2].

This mechanism is relevant because specific methylation patterns have been associated with early-life social experiences. Family psychosocial adversity has been associated with increased or decreased DNA methylation at distinct sets of genomic CpG sites [3]. Studies examining peripheral DNA from blood or saliva in infants, adolescents, and adults have also reported associations between childhood abuse or neglect and increased *NR3C1* methylation. The *NR3C1* gene is involved in regulation of the body's stress response, providing evidence that epigenetic regulation may represent one biological mechanism through which experiences of adversity become biologically embedded [4].

This raises the question of which specific genes may be associated with adversity and could also be relevant to reward-learning. One potential candidate is *SLC6A4*.

The *SLC6A4* gene encodes the serotonin transporter, which is responsible for the reuptake of serotonin from the synaptic space into presynaptic neurons [5]. Serotonin signaling plays an important role in regulating mood, emotion, motivation, and behavior. Serotonergic neurons have also been associated with reward-learning [6]. Research has shown that serotonergic neurons can respond to punishment and that a subset of these neurons is activated by reward-predicting cues, with different patterns of signaling occurring across different stages of reward and punishment processing.

Because *SLC6A4* regulates serotonin reuptake [7], changes in *SLC6A4* expression could potentially alter serotonergic signaling and, in turn, influence reward-related processes. However, this does not establish that *SLC6A4* methylation directly causes changes in reward-learning. Instead, it provides a biological reason for investigating whether the two processes are associated.

SLC6A4 methylation has also emerged as a potential marker of adversity exposure [8]. Regarding *SLC6A4*, altered methylation is relevant because serotonin influences numerous behavioral processes associated with reward and motivation, including, for example, impulsivity and goal-directed behavior. Research involving serotonergic manipulation has further demonstrated effects on goal-directed versus habitual behavior, while several serotonin receptor subtypes, including 1A, 1B, 2A, 2C, and 3, have also been involved in reward-related behavior and reward valuation [9], [10], [11], [12].

These findings provide a basis for investigating whether adversity-associated *SLC6A4* DNA methylation may be related to altered reward-learning during adolescence. Although childhood adversity has been associated independently with altered reward-learning and changes in DNA methylation, and additionally *SLC6A4* methylation has been looked into as a potential marker of adversity exposure, the potential relationship and pathway between adversity-associated *SLC6A4* methylation and altered reward-learning during adolescence remains in question.

Research question & Hypothesis

The proposed research question is:

Could *SLC6A4* DNA methylation represent a biological pathway linking childhood adversity to altered reward-learning during adolescence?

Hypothesis: Childhood adversity-associated *SLC6A4* DNA methylation will be associated with altered reward-learning during adolescence.

Proposed Study & Methodology

The proposed research study would follow a qualitative and quantitative framework with a large sample size and standardized, repeatable structure.

Childhood Adversity: Adolescents aged 12-17 would be subjected to questioning to assess level of adversity. Questions would assess varying levels, types, and chronicity of adversity. Participants would be categorized into three groups based on the severity and characteristics of reported adversity. These categories would be determined for research purposes and would not be disclosed to participants, avoiding the assignment of possibly stigmatizing labels. The questionnaire would provide a quantitative measure of adversity that could be compared with biological and behavioral measures.

***SLC6A4* methylation:** Participants would provide a saliva or blood sample to assess DNA methylation at selected CpG sites within the *SLC6A4* gene. A methylation assay, such as bisulfite pyrosequencing, would be used to treat genomic DNA and quantify methylation at the selected sites. This would provide a quantitative measure of the percentage of DNA methylation at the

targeted regions. Peripheral methylation should be interpreted cautiously, as methylation measured in blood or saliva is not a direct measure of methylation throughout the brain.

Reward-Learning: Reward-learning would be measured using a validated probabilistic reinforcement-learning task consisting of repeated trials in which participants choose between two symbols and learn through trial and error which symbol has a higher probability of producing a reward. After each choice, the participant will be provided either a *reward*—positive feedback—or a *punishment*—negative feedback. The study will also include a reversal phase where previously learned reward circumstances are changed, analyzing how swiftly a participant can *adjust* their prior notion and expectations. This test uses a model of positive and negative feedback to test individuals' ability to learn reward-outcome associations and update those associations following changes in outcome based on:

- **Learning rate** -> how quickly behavior is changed after feedback
- **Reward sensitivity** -> how strongly positive feedback influences subsequent choice
- **Punishment sensitivity** -> how strongly negative feedback influences subsequent choice
- **Reversal learning** -> what happens when the previously rewarded choice stops being rewarded
- **Choice accuracy** -> whether they increasingly choose the option associated with reward

Together, these three components would provide a combination of biological, behavioral, and self-reported data for examining the relationship between childhood adversity, *SLC6A4* methylation, and reward-learning.

Relevance

Scientific: This study addresses the intersection of epigenetics, gene regulation, and the biological effects of childhood adversity. By examining *SLC6A4* DNA methylation alongside reward-learning behavior in adolescents, the proposed research could provide insight into whether epigenetic variation may be associated with differences in reward-related learning following experiences of adversity.

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Clinical Research: The proposed study focuses on the *SLC6A4* gene, which encodes the serotonin transporter and plays an important role in serotonin regulation [5]. Understanding the relationship between adversity, *SLC6A4* methylation, and reward-learning may contribute to broader research on emotional regulation, reward-related behavior, and serotonergic function.

Psychological: By combining self-reported adversity, a biological measure of *SLC6A4* methylation, and behavioral measures of reward-learning, the study seeks to connect environmental experiences with biological and psychological processes involved in decision-making.

Limitations & Project Logistics

This project would require informed consent and signatures from both adolescents and their parents or legal guardians, creating additional concerns associated with conducting research involving minors. The study would require a considerably large sample size to examine associations among childhood adversity, *SLC6A4* methylation, and reward-learning while accounting for variation in age and demographic characteristics.

The study would also be subject to limitations associated with self-report. Participants may lie, interpret questions differently, or may be unwilling to fully report sensitive experiences, creating the possibility of response or recall bias. Because the study is observational, associations between adversity, *SLC6A4* methylation, and reward-learning would not necessarily establish causation. Additionally, methylation measured from peripheral samples such as saliva or blood would not directly represent the exact type of methylation occurring in the brain.

Ethical Implications

Since the proposed study involves minors and questions about potentially sensitive experiences of childhood adversity, participant safety and privacy would be essential. Participation would require informed consent from a parent or legal guardian as well as consent from the adolescent. Participants would be informed that participation is in fact voluntary and that they may decline to answer some questions or withdraw from the study.

Personal data and biological information would have to be kept confidential and stored using appropriate security procedures. Researchers would also need to determine appropriate procedures prior to the study for responding to disclosures of current abuse or immediate safety concerns, aligning procedures with applicable research and mandatory-reporting requirements.

Because the study involves biological samples, participants and guardians would also need to be informed on what is being collected, how samples would be analyzed, how long data would be retained, and how biological information would be protected and used.

Conclusion

In conclusion, this research study investigates associations among childhood adversity, *SLC6A4* methylation and reward-learning in adolescents. By examining these measures together, the study seeks to determine whether differences in *SLC6A4* methylation are associated with the relationship between childhood adversity and altered reward-learning.

Timeline

Week 1: Literature review — Review existing research on childhood adversity, gene methylation, and reward-learning.

Week 2: Study design and preparation — Plan and finalize the participant criteria, the adversity level assessment, the *SLC6A4* methylation analysis technique, and the reward-learning task.

Week 3: Data collection — Recruit adolescent participants, extract DNA samples for methylation analysis, introduce privacy safeguards, collect necessary consent waivers, and administer reward-learning tests and adversity questionnaires.

Week 4: Data analysis — Organize data and analyze associations from data on childhood adversity, *SLC6A4* methylation, and observed reward-learning.

Week 5: Study write up and conclusion — Consider study limitations, synthesize and interpret findings, evaluate proposed research question and hypothesis, and finalize research paper.

Post Project Plan

Following the proposed research study, future research should further investigate:

- Biological pathways connecting *SLC6A4* gene expression, serotonergic function, and reward-learning
- Relationship between childhood adversity and altered reward-learning

Future research should also utilize larger and more diverse samples. It should additionally incorporate direct measures of serotonergic function and serotonin production to investigate if the associations shown in this study could represent a broader, more connected mechanism.

Tab 3

Abstract: Childhood adversity has been researched and shown to be associated with altered reward-learning. Increased exposure to adversity may influence adolescent reward-learning by increasing sensitivity to negative feedback. However, gaps in existing research remain regarding the biological mechanisms underlying the relationship between adversity and altered reward-learning. One potential pathway through which adversity may influence learning is epigenetic regulation. Environmental stressors have been associated with changes in DNA methylation, which can influence gene expression. *SLC6A4*, a gene encoding the serotonin transporter, may represent a potential biological link between adversity and altered reward-learning. This study investigates *SLC6A4* DNA methylation as a potential link and hypothesizes that childhood adversity-associated *SLC6A4* DNA methylation will be associated with altered reward-learning in adolescents. The proposed study will assess *SLC6A4* DNA methylation, identify varying levels of childhood adversity, and measure reward-learning using a probabilistic behavioral task with a reversal component.

Introduction: Adversity is broadly defined as exposure to distressing or harmful life circumstances, ranging from abuse to neglect. In this study, adversity specifically refers to early-life stress (ELS) associated with unstable and abusive caregiving environments. Childhood adversity has been associated with differences in how individuals learn from rewards, risks, and feedback. Reward-learning involves an individual's ability to recognize signals in the environment and associate them with rewards or risks. This process contributes to decision-making by allowing individuals to modify future behavior based on positive or negative feedback [1]. Importantly, adverse experiences do not necessarily prevent individuals from recognizing associations between events and rewarding outcomes. Instead, adversity may alter how individuals assign motivational value to this information and incorporate it into future behavioral decisions. Although the relationship between adversity and altered reward-learning has been increasingly studied, the mechanisms beneath this relationship remain not entirely understood. Children exposed to adverse experiences may recognize standard positive outcomes but prioritize avoidance of potential risks, devalue information about potential rewards, present greater sensitivity to negative than positive feedback, demonstrate increased sensitivity to reversals from positive to negative outcomes than from negative to positive outcomes, and exhibit altered activity in reward-related neural regions during reward anticipation and reception [1]. One proposed mechanism through which environmental experiences may become biologically embedded is through epigenetic changes.

Literature Review: Epigenetics refers to biological mechanisms that regulate gene expression without changing the underlying DNA sequence. One mechanism is DNA methylation, which could potentially contribute to the biological effects of adversity on behavior. DNA methylation occurs when a methyl group is added to cytosine-guanine sites on DNA, known as CpG sites [2]. Depending on its location and biological context, DNA methylation can influence whether a gene is expressed [2].

This mechanism is relevant because specific methylation patterns have been associated with early-life social experiences. Family psychosocial adversity has been associated with increased or decreased DNA methylation at distinct sets of genomic CpG sites [3]. Studies examining peripheral DNA from blood or saliva in infants, adolescents, and adults have also reported

associations between childhood abuse or neglect and increased *NR3C1* methylation. The *NR3C1* gene is involved in regulation of the body's stress response, providing evidence that epigenetic regulation may represent one biological mechanism through which experiences of adversity become biologically embedded [4].

The *SLC6A4* gene encodes the serotonin transporter, which is responsible for the reuptake of serotonin from the synaptic space into presynaptic neurons [5]. Serotonin signaling plays an important role in regulating mood, emotion, motivation, and behavior. Serotonergic neurons have also been associated with reward-learning [6]. Research has shown that serotonergic neurons can respond to punishment and that a subset of these neurons is activated by reward-predicting cues, with different patterns of signaling occurring across different stages of reward and punishment processing. Because *SLC6A4* regulates serotonin reuptake [7], changes in *SLC6A4* expression could potentially alter serotonergic signaling and, in turn, influence reward-related processes. However, this does not establish that *SLC6A4* methylation directly causes changes in reward-learning. Instead, it provides a biological reason for investigating whether the two processes are associated.

Regarding *SLC6A4*, altered methylation is relevant because serotonin influences numerous behavioral processes associated with reward and motivation, including, for example, impulsivity and goal-directed behavior. Research involving serotonergic manipulation has further demonstrated effects on goal-directed versus habitual behavior, while several serotonin receptor subtypes, including 1A, 1B, 2A, 2C, and 3, have also been involved in reward-related behavior and reward valuation [9], [10], [11], [12].

Research question & Hypothesis — The proposed research question is:

Could *SLC6A4* DNA methylation represent a biological pathway linking childhood adversity to altered reward-learning during adolescence?

Hypothesis: Childhood adversity-associated *SLC6A4* DNA methylation will be associated with altered reward-learning during adolescence.

Proposed Study & Methodology

Childhood Adversity: Adolescents aged 12-17 would be subjected to questioning to assess level of adversity. Questions would assess varying levels, types, and chronicity of adversity. Participants would be categorized into three groups based on the severity and characteristics of reported adversity. These categories would be determined for research purposes and would not be disclosed to participants, avoiding the assignment of possibly stigmatizing labels. The questionnaire would provide a quantitative measure of adversity that could be compared with biological and behavioral measures.

***SLC6A4* methylation:** Participants would provide a saliva or blood sample to assess DNA methylation at selected CpG sites within the *SLC6A4* gene. A methylation assay, such as bisulfite pyrosequencing, would be used to treat genomic DNA and quantify methylation at the selected sites. This would provide a quantitative measure of the percentage of DNA methylation at the

targeted regions. Peripheral methylation should be interpreted cautiously, as methylation measured in blood or saliva is not a direct measure of methylation throughout the brain.

Reward-Learning: Reward-learning would be measured using a validated probabilistic reinforcement-learning task consisting of repeated trials in which participants choose between two symbols and learn through trial and error which symbol has a higher probability of producing a reward. After each choice, the participant will be provided either a *reward*—positive feedback—or a *punishment*—negative feedback. The study will also include a reversal phase where previously learned reward circumstances are changed, analyzing how swiftly a participant can *adjust* their prior notion and expectations. This test uses a model of positive and negative feedback to test individuals' ability to learn reward-outcome associations and update those associations following changes in outcome.

Together, these three components would provide a combination of biological, behavioral, and self-reported data for examining the relationship between childhood adversity, *SLC6A4* methylation, and reward-learning.

Relevance

Scientific & Clinical: This study addresses the intersection of epigenetics, gene regulation, and the biological effects of childhood adversity. By examining *SLC6A4* DNA methylation alongside reward-learning behavior in adolescents, the proposed research could provide insight into whether epigenetic variation may be associated with differences in reward-related learning following experiences of adversity. Because the serotonin transporter is also a major target of selective serotonin reuptake inhibitors (SSRIs), research examining *SLC6A4* and its regulation may potentially have relevance to future research on serotonergic treatments [5], [10].

Psychological: The findings could help clarify potential pathways connecting childhood adversity with differences in reward-related behavior during adolescence.

Limitations & Project Logistics: This project would require informed consent from adolescents and their parents or legal guardians. Self-report may introduce response or recall bias. Because the study is observational, associations between adversity, *SLC6A4* methylation, and reward-learning would not establish causation. Additionally, peripheral methylation from saliva or blood would not directly represent methylation occurring in the brain.

Ethical Implications: Since the study involves minors and sensitive experiences, participant safety and privacy would be essential. Participation would require informed consent, confidentiality, and the ability to decline questions or withdraw. Researchers would also need procedures for responding to disclosures of current abuse or immediate safety concerns in accordance with applicable mandatory-reporting requirements.

Conclusion: In conclusion, this research study investigates associations among childhood adversity, *SLC6A4* methylation and reward-learning in adolescents. By examining these measures together, the study seeks to determine whether differences in *SLC6A4* methylation are associated with the relationship between childhood adversity and altered reward-learning. The

study is expected to find differences in both reward-learning and gene methylation correlated with varying levels of adversity, filling an evident gap in mechanism and pathway research of *SLC6A4* methylation, its biological connection to decision-making and adaptability, and adversity's effects on both subject areas. In particular, the study could provide a basis for further investigation into the relationship between *SLC6A4* regulation, serotonin-related processes, and decision-making. This research has potential relevance to biological, psychological, and clinical research. It could identify patterns and associations that inform future studies examining how environmental experiences may become biologically embedded and relate to behavioral development.

Timeline: Week 1: Literature review — Review existing research on childhood adversity, gene methylation, and reward-learning; **week 2:** Study design and preparation — Plan and finalize the participant criteria, the adversity level assessment, the *SLC6A4* methylation analysis technique, and the reward-learning task; **week 3:** Data collection — Recruit adolescent participants, extract DNA samples for methylation analysis, introduce privacy safeguards, collect necessary consent waivers, and administer reward-learning tests and adversity questionnaires; **week 4:** Data analysis — Organize data and analyze associations from data on childhood adversity, *SLC6A4* methylation, and observed reward-learning; **week 5:** Study write up and conclusion — Consider study limitations, synthesize and interpret findings, evaluate proposed research question and hypothesis, and finalize research paper.

Post Project Plan: Following the proposed research study, future research should further investigate biological pathways connecting *SLC6A4* gene expression, serotonergic function, and reward-learning, detailed psychological implications of altered *SLC6A4* gene expression, and the relationship between childhood adversity and altered reward-learning. Future research should also utilize larger and more diverse samples. It should additionally incorporate direct measures of serotonergic function and serotonin production to investigate if the associations shown in this study could represent a broader, more connected mechanism.